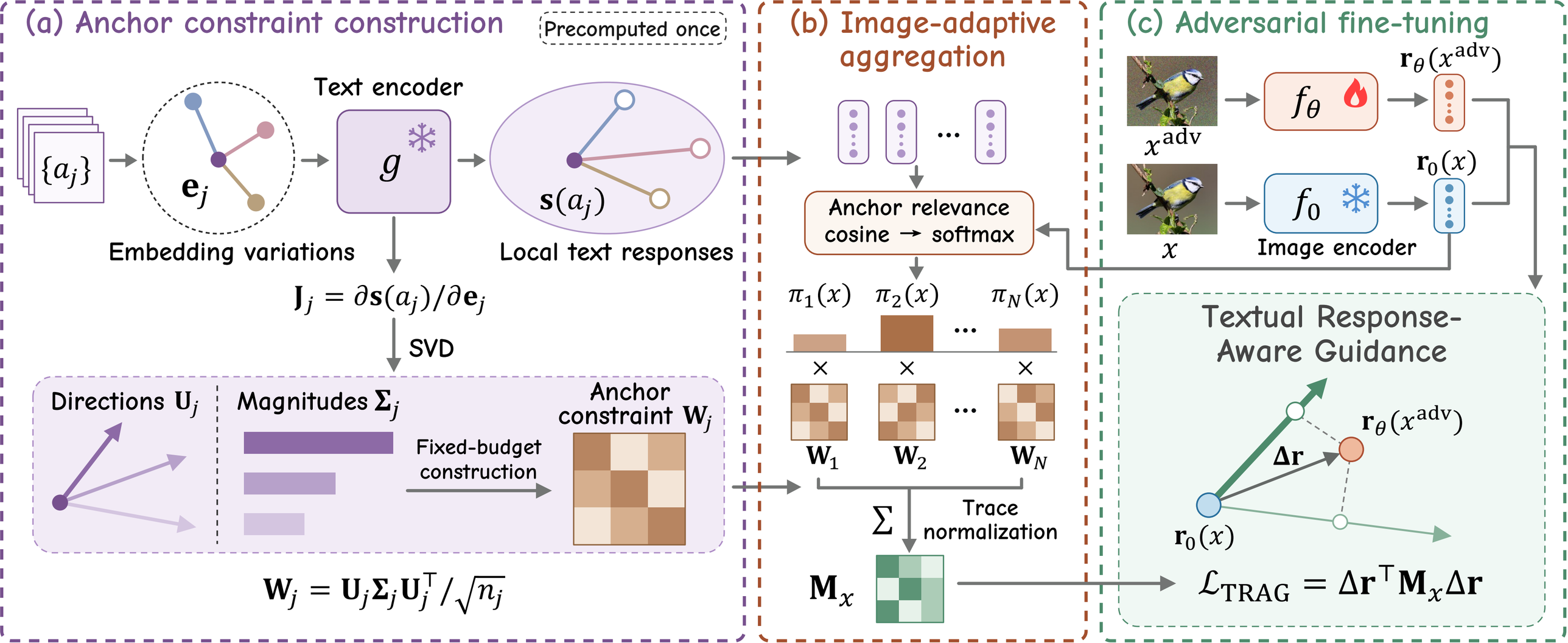

(a) Anchor constraint construction
Precomputed once
Text encoder
$\{a_j\}$
$\mathbf{e}_j$
g
$\mathbf{s}(a_j)$
Embedding variations
Local text responses
$\mathbf{J}_j = \partial \mathbf{s}(a_j)/\partial \mathbf{e}_j$
SVD
Directions $\mathbf{U}_j$
Magnitudes $\mathbf{\Sigma}_j$
Fixed-budget construction
Anchor constraint $\mathbf{W}_j$
$\mathbf{W}_j = \mathbf{U}_j \mathbf{\Sigma}_j \mathbf{U}_j^\top / \sqrt{n_j}$
(b) Image-adaptive aggregation
Anchor relevance cosine → softmax
$\pi_1(x)$ $\pi_2(x)$ $\pi_N(x)$
$\mathbf{W}_1$ $\mathbf{W}_2$ $\mathbf{W}_N$
Σ
Trace normalization
$\mathbf{M}_x$
(c) Adversarial fine-tuning
x^{adv}
f_θ
$\mathbf{r}_\theta(x^{\text{adv}})$
x
f_0
Image encoder
$\mathbf{r}_0(x)$
Textual Response-Aware Guidance
$\mathbf{r}_\theta(x^{\text{adv}})$
$\Delta\mathbf{r}$
$\mathbf{r}_0(x)$
$\mathcal{L}_{\text{TRAG}} = \Delta\mathbf{r}^\top \mathbf{M}_x \Delta\mathbf{r}$